UBC Department of Computer Science CPSC 404: Advanced Relational Databases Course Outline (Syllabus) for September-December 2025

1 Course Description

Whereas CPSC 304 focuses on *using* a relational database management system (RDBMS), CPSC 404 focuses on the *internals* and *performance issues* of an RDBMS. Topics include: disks and the storage hierarchy, the I/O cost model, buffer pool management, disk scheduling, record layouts, metadata, system catalogs, tree-structured indexes (especially B+ trees), hash indexes, query evaluation, query optimization, transaction processing, concurrency, and crash recovery.

2 Prerequisites

The prerequisites are CPSC 304 and one of CPSC 213, CPSC 261, or CPEN 212. Equivalent courses from other institutions that map to these are acceptable. Although Workday prevents most invalid registrations, our department also does prerequisite-checks to detect any registrations that fell through the cracks. Students without appropriate prerequisites will get a "missing prerequisite" letter near the start of the course. Follow the instructions in the letter to resolve it. If you do not have the prerequisites, or if you already have credit for a CPSC 404-like course, then you will be dropped from the course.

3 Basic Information

3.1 Lectures

Tuesday/Thursday 3:30PM to 5:00PM (Pacific) in SWNG 222 or on Panopto (streaming and recording). There are no labs or tutorials for this course.

3.2 Office Hours

Online using Zoom or in person. The times will be announced shortly after the start of the term and will also be listed on Canvas.

4 Communication

E-mail and E-mail Addresses: Because this is a very large class, it is simply not practical for the instructor or the TAs to respond to individual e-mail requests (or private postings) of a general nature. Limit such e-mail to items having a personal or confidential nature (e.g., prolonged illness). Lectures, office hours, and Piazza are suitable places to ask general questions. You can find the contact information for the teaching team on Canvas.

Piazza Communication on Exam Days: To encourage appropriate study habits, we will stop holding office hours/ answering questions on the discussion board 24 hours before each exam.

5 Covid Safety

If you're sick, it's important that you stay home – no matter what you think you may be sick with (e.g., cold, flu, other).

Do not come to class if you are sick, have recently tested positive for Covid, or are required to stay home. This precaution will help reduce risk and keep everyone safer. In this class, the marking scheme is intended to provide flexibility so that you can prioritize your health and still be able to succeed as outlined throughout, including planning to stream all classes and put all recordings on Canvas.

If I (the instructor) am sick: I will do my best to stay well, but if I am ill, then I will not come to class. If that happens and I am well enough to teach, then I will teach via Zoom. If not, then class will be cancelled. In either case, I will inform you via Piazza as soon as possible.

Please consider wearing a mask and getting vaccinated: http://www.vch.ca/covid-19/covid-19-vaccine#clinics. The higher the rate of vaccination in our community overall, the lower the chance of spreading this virus. You are an important part of the UBC community. Please arrange to get vaccinated if you have not already done so.

6 Waitlist

If you are on the waitlist, then you need to attend class, and you should submit all of the in-class exercises while waiting. Submit your clicker responses using iClicker Cloud (the normal way) during class. To submit "in-class" exercises while you are on the waitlist, please refer to the instructions on Canvas. We won't have any in-class exercises or clicker questions that count for marks during the first lecture.

A detailed explanation of how the Department of Computer Science manages its waitlists is at https://www.cs.ubc.ca/students/undergrad/courses/waitlists

7 Textbook/Clickers

7.1 Textbook

Raghu Ramakrishnan and Johannes Gehrke. *Database Management Systems*, 3rd Edition, McGraw-Hill, 2003. ISBN 9780072465631. You can buy physical copies at the UBC bookstore or wherever textbooks are sold. You can also get a digital copy at the bookstore online: https://bookstore.ubc.ca/. There may be both a full version (the more expensive option) and a partial e-textbook available on the bookstore website that covers only the chapters for CPSC 404 (the cheaper option). If you buy the partial version, try it early, since I had trouble with it!

Copies of the hardcover textbook are on reserve for CPSC 304 and 404 at the CS Reading Room and at the UBC library.

7.2 Clicker

We will use **iClicker Cloud** (also known as iClicker REEF). It allows you to respond to clicker questions with either a mobile device or a computer. It is free. Here are the instructions for students. When you register for iClicker Cloud, and you connect your registration to Canvas, then Canvas will automatically pick up your scores. You are responsible for making sure that you complete this integration step properly!

8 Evaluation

The final grades will ***tentatively*** be calculated as follows. We reserve the right to change this grading scheme in case some unforeseen events come up beyond the instructors' control. If we do make a change, we will certainly inform you.

Component	Weighting
"In-Class" Exercises	10%
Clickers	4%
SQL Server DBA Exercise	10%
Midterms (2) – each weighted equally	36%
Final Exam (Date to be announced!)	40%

8.1 Minimum Passing Criteria

To pass the course, you must obtain a 50% overall mark, and get an overall passing grade on the combined (weighted) midterm/final exam portion of the course. In keeping with department policies, students who do not pass the midterm/final exam portion will receive the minimum of 45% and their overall failing grade.

8.2 "In-Class" Exercises

The "in-class" exercises will be assigned during the scheduled class times. We'll take the best 80% of the days to allow for the odd missed class. The in-class exercises will be marked on a 0/1/2 point scale for effort and completion—not necessarily for the correct answers. These will be handed in, on Canvas, by 10PM the day after the date that an in-class exercise is started in class. We will not accept late submissions for in-class exercises, since we will be posting solutions to them. It is your responsibility to ensure you uploaded the correct file for submission. We will only grade the last submission of an assignment, and we will not look in other places on Canvas for your submission.

8.3 Clickers

Clickers will be graded for participation only. As with the in-class exercises, we'll take the best 80% of the points to allow for the odd missed class and/or technical difficulties. Be aware that the streaming service often is delayed by up to two minutes. Thus, streaming the course and using clickers will probably not work well. With very limited exceptions, we will ***not*** be excusing missed Clicker questions other than the above, since that is already accounted for. Note: saying that the system failed to record your answers to a set of questions is not an exceptional circumstance that merits an exception.

If your in-class exercise grade is higher, or, if for some reason, clickers do not work for the professor this term, we will use that for your clicker grade instead.

8.4 SQL Server DBA Exercise

Part of your grade depends on doing the SQL Server exercise, which will be given out in November. You can either use the PCs in a CPSC undergraduate lab or use our **remote lab** environment (without setting foot in a lab). In both cases, you will access the SQL Server production instance (sometimes called the "database engine" or the "back end" database server) hosted by our department using the `mayne` server.

8.5 Examinations

Each exam will be on paper, closed book and closed notes ***with a simple non-programmable (non-graphing) calculator allowed***; exams are to be completed individually. You are NOT permitted to access any of the course materials, including your notes, during the exam. You are also NOT to communicate with anyone about the exam during the scheduled write time – you are to work independently. Communication with other students (written, text, verbal, etc.) is not permitted and will constitute Academic Misconduct. If you violate these conditions you have engaged in Academic Misconduct and will be subject to the consequences as determined by the university.

8.5.1 Midterm examinations

There will be two midterms. However, we will automatically replace your worst/missing midterm (but only one!) with your final exam mark if your final exam mark is higher. We're doing this because there may be issues beyond your control including COVID challenges, sickness, personal situations, stress, really bad day, road closure, etc.

If you have an unplanned absence (e.g., due to illness) for a midterm, you should provide either documentation or fill out a self-declaration form; your documentation should be emailed to your instructor or posted in a private piazza post. However, note that for a second or subsequent request for academic concessions resulting from acute illness, we will refer you to your academic advising office, graduate supervisor, or graduate advisor. The final covers more topics than the midterms.

The midterms will be held DURING CLASS TIME on:

Midterm	Date
1	Tuesday 2025/10/21
2	Tuesday 2025/11/18

Because the midterms will be held during class time, there will be no makeup exams. Please see above for what happens if you miss a midterm.

8.5.2 Final Examination

The final examination will be written on a date in December to be determined by the University. Please do not schedule any travel plans prior to the announcement of the final exam date; travel is not an accepted reason for the delay or deferment of a final exam. If you cannot write the final exam, then you must provide documentation (e.g., a doctor's note) to your home faculty office within 48 hours of missing the exam. For example, if you are in Science, then it will be the Faculty of Science; if you are in Engineering, then it will be the Faculty of Applied Science; and similarly for Arts, Commerce, etc. Assuming the faculty advising office approves of the deferment request, you will write an exam with the subsequent course offering.

Important note: Your home faculty office may not allow you to write a deferred final exam (even if you have a doctor's note) if your term work is incomplete (e.g., missing midterm, missing classes "in-class" exercises).

Learn more and find the application online:
https://science.ubc.ca/students/advising/concession

8.6 Regrading

Unless a different deadline is explicitly given, regrade requests will be accepted up to one week after the grades are released. When a regrade request is submitted, we reserve the right to regrade the whole deliverable/exam – not just a portion of it. You can submit non-exam related regrade requests here; keep an eye on Piazza for instructions on how to submit a regrade request for an exam.

Things to keep in mind about regrade requests:

- A regrade request is **not** an opportunity for you to argue about the grading scheme. It is a chance for you to address situations where the grading scheme was applied incorrectly to your work. Regrade requests that debate the validity of the grading scheme (e.g., this item should not be worth this number of points or this item should not be included in the rubric) will be discarded without warning.
- Issues already discussed on Piazza and for which the course team has declared that the grading stands should not be brought up again in a regrade request.
- The regraded mark is final– even when the regraded mark is less than the original grade. You cannot choose to withdraw a regrade request.
- If we receive multiple regrade requests for an assessment, we will only consider the very last request that was submitted.

- When submitting a regrade request, we will ask you for an explanation of why you feel the need to submit the request. We are not looking for an essay. We want to have a meaningful discussion with you about why particular elements of your answer deserve to be awarded points based on the grading scheme. Unless the grading error is very obvious, it is generally in your best interest to spend some time on answering this part of the regrade request.
- Regrade requests received with reasons that do not discuss the answer with relation to the grading scheme (e.g., "I tried really hard and this grade does not reflect my effort" or "I don't agree") may be discarded without warning.
- Late regrade requests are not accepted.
- Incomplete regrade requests will be discarded.
- We will only start processing regrade requests after submission deadline has passed.
- Regrade requests submitted via the incorrect avenue (e.g., through email) will not be considered.
- Regrade requests that violate these rules are subject to a 5% additional penalty. The deduction is calculated based on the full value of the item being regraded.

9 Electronic Information Sources

9.1 Canvas

Lecture notes, recordings, sample assignment questions (with full solutions), and exam practice questions (with full solutions) will be available on Canvas. You are responsible for all material presented in the lectures—not just what's written on the lecture slides—as well as all related material on Canvas (including the corresponding textbook readings, in-class exercises, and practice questions). I will post preliminary slides before lecture (which will not include things like clicker questions) and final annotated slides (including clicker questions and answers) after each lecture.

UBC's ITServices group hosts Canvas. If you don't already have a Campus-Wide Login (CWL) account, you should visit `www.cwl.ubc.ca` to get one. All students registered in CPSC 404, and who have a CWL ID, will automatically have their CWL ID linked to the CPSC 404 Canvas pages. If you are taking other courses that use Canvas, then your CWL ID will be linked to those courses, as well.

We will make the Canvas site available to all UBC members until after the drop without a W deadline to make it accessible to those who are on the waitlist.

9.2 Piazza

Piazza will host our course discussion board. It is ***required reading*** for this course. You should read it at least once per day. The TAs and instructor will be monitoring and responding to questions daily, but students are also encouraged to respond to each other's postings and questions. Do not post code, solutions, or copyrighted information on the discussion board (other than perhaps small fragments, if necessary).

Questions about the CPSC 404 course contents (e.g., lecture, textbook, exercises, etc.) can be posted on Piazza, *but please check to make sure that your question hasn't already been asked or answered*. In previous terms, students have asked the same questions over and over. Remember, you can always re-read Piazza notes that you've already seen, and you can search for keywords within Piazza. Students who are posting notes will be anonymous to other students, but not to the instructional staff.

Problems with Canvas itself (i.e., other than with the content of the CPSC 404 web pages) should be directed to help@itservices.ubc.ca. Problems with undergrad accounts, servers, or hardware problems in the lab should be directed to help@ugrad.cs.ubc.ca.

You can access Piazza's CPSC 404 discussion board via Canvas. On Canvas, there is a link, instructions, and the **access code**. I'll also mention it in class.

Alternatively, the class can be accessed via this link.

Piazza registration instructions are found on Canvas – click on the Piazza tab on the LHS. Piazza does *not* automatically use your Canvas-associated e-mail as your Piazza e-mail. When signing up for Piazza for the first time, you can choose what e-mail address to use, and you can change it anytime.

- First, a UBC legal/privacy disclaimer: "If you decide to use Piazza in your course, you are required to use the Canvas integration [done] to ensure that its use is compliant with the Freedom of Information and Protection of Privacy Act (FIPPA). The Canvas integration link directs students to the proper Piazza course and masks the student's identity before it is sent to the site. To comply with BC privacy legislation, students will need to create a Piazza account and agree to the terms of use the first time they use the tool. While Piazza adheres to strict U.S. privacy regulations (FERPA), UBC cannot guarantee the security of student's private details on servers outside of Canada. Students should be reminded to exercise caution whenever using personal information, and that they may use a pseudonym to protect their privacy if they have concerns."

When signing up for Piazza, use whichever email address makes you the most comfortable (whether that is your cs_id@students.cs.ubc.ca) or a something else. If you choose an email with a pseudonym, please keep it school appropriate and don't pick the name of a celebrity, sports star, other student, etc. You will need to ensure that your account is associated with a UBC affiliated email address (this can be one or more of your ugrad.cs.ubc.ca, students.cs.ubc.ca, student.ubc.ca email addresses). You will find the link to the Piazza course on Canvas.

9.3 Panopto

Lecture will be streamed and recorded via Panopto. The recordings are released by UBC IT services, and we cannot control the timing. Disclaimer: If the technology fails (e.g., the stream stops halfway during class, the recording fails etc.), a new recording may **not** be made to replace whatever is missing.

10 Course Topics and Learning Goals (Tentative)

Introduction	Course overview, DB-related tasks, self-managing DBMSs, definitions of DB2 database objects
Storage (Chapter 9)	Storage/memory hierarchy, disks, access times, I/O cost model (number of page I/Os), buffer pool management, disk scheduling, metadata (system catalogs, including IBM's DB2 catalog pages), record and page layouts (fixed and variable), free space and free page calculations
Tree-Structured Indexes (Chapter 10 and some of Chapter 8)	Introduction to indexing; clustering; B+-trees for searching, inserting, updating, and deleting; I/O costs; complexity analysis
Hash-Structured Indexes (Chapter 11)	Extendible hashing, linear hashing; analysis
External Sorting (Chapter 13)	External sorting strategies and analysis, including cylindrification and large block sizes
Query Evaluation (Chapters 12 & 14)	Relational algebra, SQL, join algorithms, buffer sizes, aggregation (group-by operations), access paths, index usage
Query Optimization (more from Chapters 12 & 14)	Query evaluation strategies for efficiently executing a database query; the role of the optimizer in DB performance; pipelining
Transaction Management and Concurrency Control (Chapters 16 & 17)	ACID properties, concurrency, serializability, isolation levels, deadlock (including deadlock detection and prevention algorithms), concurrency control (e,g,, 2PL, Strict 2PL, Multiple-Granularity locks, index locking)
Logging and Crash Recovery (Chapter 18)	Backup and recovery (application-level and system-level), ARIES algorithm, logging, checkpoints

The course-level and topic-level **learning goals** are provided on Canvas. The lecture slides will also contain the topic-level learning goals.

11 Collaboration/Cheating Policies (Academic Misconduct)

Here is the University Policy on Academic Integrity:

"Academic honesty is essential to the continued functioning of the University of British Columbia as an institution of higher learning and research. All UBC students are expected to behave as honest and responsible members of an academic community. Breach of those expectations or failure to follow the appropriate policies, principles, rules, and guidelines of the University with respect to academic honesty may result in disciplinary action."

A more detailed description of academic integrity, including the University's policies and procedures, including links to other campus-wide policies and procedures can be found in the UBC Calendar at:

https://vancouver.calendar.ubc.ca/campus-wide-policies-and-regulations/academic-honesty-and-standards

Important note: You may NOT collaborate at all on the midterms, the final exam, or any work that we explicitly state must be done individually. Your work must be your own. You cannot use a friend or tutor to do any of your work. If, due to a COVID emergency, we have online exams, you cannot share your screen content (or exam questions or answers) in any way.

The use of generative artificial intelligence tools is strictly prohibited in all course assignments unless explicitly stated otherwise. This includes ChatGPT and other artificial intelligence tools and programs.

Don't look for loopholes to cheating. Use common sense and don't cheat. Penalties include getting zero for the course, and suspension from the University.

12 Other Important Notes and Links

In the event of extreme weather conditions, power outages, environment emergencies, etc., we may need to move a class to online mode (or provide a video recording) … or an exam may need to be rescheduled. Be aware that the dates listed in this course outline are tentative. We will work with the University, including the Faculty of Science's Information Centre and Dean's office, to determine practical solutions.

Indigenous Land

UBC Vancouver is located on the traditional, ancestral, and unceded territory of the Musqueam people. The land is situated on, and has always been, a place of learning for the Musqueam, who for millennia have passed on their culture, history, and traditions from one generation to the next on this site. See the UBC Indigenous Portal for more details: https://indigenous.ubc.ca/indigenous-engagement/

Equity & Inclusion

We aim to build a community where equity and inclusion are embedded in all aspects of campus life. If you require assistance related to issues of equity, discrimination or harassment please contact the Equity & Inclusion Office:

Brock Hall, Room 2306
604-822-6353
info@equity.ubc.ca
https://equity.ubc.ca

Health & Wellness

Health and wellness, both physical and mental, are important for academic success. If you are having difficulty with your studies, or feel overwhelmed, or are experiencing distress, UBC provides a number of resources to help, including 24/7 crisis support. You can also find tips on how to approach a friend who may be experiencing difficulties. For more information, please visit:

https://science.ubc.ca/students/wellbeing

If you are seriously ill or are dealing with a significant issue (e.g., the death of a close family member) that may prevent you from performing well in your courses, please contact your instructor and/or your home faculty office, as soon as possible.

If you face similar circumstances for your final exam, please talk to your home faculty's advising office *before* you take the exam. They will assist you in determining an appropriate course of action. Please note that different Faculties handle requests for academic concession in different ways. Links to further information for the Faculties of Arts, Science and Commerce are provided below:

Science: Exam Issues
Sauder: Academic Concession
Arts: Academic Concession

Centre for Accessibility

The Centre for Accessibility provides support for students with disabilities, chronic medical conditions, and various other challenges. If this applies to you and you require academic accommodations to meet course objectives, please contact the Centre for Accessibility:

Brock Hall, Room 1203
604-822-5844
info.accessibility@ubc.ca
https://students.ubc.ca/about-student-services/centre-for-accessibility

13 Acknowledgements

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